Banking integration and deliverable readiness audit

Date: 21 October 2026. Scope: the merged repository, its connected execution path, shared regression tests, the two-page course handout and the internal project guidelines. Current user scope is banking only. See the operational guide for all commands and policy diagrams.

Budget-grid update

The current user-confirmed grid is $k \in \{0, 1, 3, 5, 7, 9\}$. Planner defaults, pilots, full-run commands, synthetic demo artifacts and the legacy pending-only plan now use this grid. Existing frozen evaluation results remain unchanged. The updated four-policy banking sweep has $16(1+9) \times 4 \times 6 \times 3 = 11520$ episodes; its held-out subset has $13(1+9) \times 4 \times 6 \times 3 = 9360$ episodes. The updated 11,520-row banking manifest was generated and every row passed offline validation; all 194 harness tests passed (one opt-in live test skipped). The refreshed synthetic artifacts contain 180 cells, and the pending-only legacy manifest contains 2,592 rows with its original schema. The earlier 9,600-row validation below records the original audit, not the new experiment size.

Current readiness

The code is ready for the documented banking evaluation workflow. Integration and multi-directory reporting are implemented. Latest validation: **233 harness tests passed, 1 opt-in live model test skipped, 30 subtests passed**. The historical discrepancy table below is retained as an audit record of resolved issues, not a list of outstanding blockers.

Run preflight and the current-version development pilot before promoting to the full evaluation. The latest `--check` call from this session could not reach the configured local Ollama endpoint. The repository contains an earlier 72-episode live pilot with all episode statuses `ok`, but those records have no semantic-policy version field and therefore do not validate the current decision-effective R3/R4 implementation. The offline smoke has 10 successful scripted episodes and remains engineering evidence only.

The user selected local `qwen3:14b` for agent, assessor and reviewer in separate contexts. Shared model weights can cause correlated failures. Current-version empirical results and confirmed trajectories remain necessary for report conclusions.

Decision-effective integration update

`hybrid_llm` and `hybrid_llm_fixed` now share decision-effective R4 processing; only their allocation threshold differs. Deterministic states and exact grants remain untouched. Live-label mode is `review_selection`; stub ablations retain `advisory`. Runtime configuration, episode results and per-call audits identify version `r3-r4-review-selection-v1`. The native exporter adds `semantic_decisions.csv`. Regression coverage includes decision-changing positive/negative findings, atomic validation, no model-created execution authority, missing task citations, protected checks, incomplete evidence, real gate review enforcement, the actual R4 adapter with a fake transport, and native semantic CSV/version validation. Validation after this integration: **225 harness tests passed, 1 opt-in live test skipped, 30 subtests passed**; the root five-test suite also passed. Local Ollama preflight remains unavailable; no live model result is claimed. The original audit table below describes the

earlier integration stage; its advisory limitation is superseded by this update. Existing run directories must not be resumed under the new source/configuration.

Discrepancies and resolutions

Finding	Resolution
Root and harness README described superseded all-domain scope and unfinished integration	Replaced active documentation with banking-only status, current commands and source precedence; retained `R3/` as historical material
Default planner selected all domains and only baselines plus metadata policy, omitting the scope-aware policy	CLI now defaults to banking, both required baselines, `hybrid_llm`, and `hybrid_llm_fixed`; task/attack-objective cohort selectors added; general library API compatibility retained
Live runner always built the scope policy with a stub despite implemented R4 code	Explicit live labels now instantiate `LocalR3Assessor` using the actual model client; live policy construction without an assessor is rejected; stub labels remain explicit
R4 output costs/statuses were absent from live usage	Per-assessment traces in `assessor.jsonl`; actual policy API attempts in `usage.jsonl`; unavailable preflight assessments do not masquerade as paid calls; local Qwen thinking disabled in the connected assessor configuration
R4 prompt content omitted from frozen source fingerprints	Fingerprints now include all `src/prompts/*.txt`, runtime Python sources, tool metadata and reviewer prompt
Legacy R5 analyzer expected `episode_results.jsonl`, `gate_decisions.jsonl`, `cost_calls.jsonl`; runner emitted native episode/audit/usage files	Added `src.banking_report`, directly consuming frozen native runner files and reusing `r3_compare` validation; legacy interfaces retained and clearly identified
Combining clean and attacked rows would invert clean security meaning and contaminate TCR	Native analysis separates condition groups; clean security sentinel never becomes attack success; attacked TCR uses official utility and attack success with a documented proxy limitation
Legacy HITL aggregation could collapse repeated task identities and used an ambiguous success cohort	Native exports aggregate consultations by unique episode; both utility-success and safe-success totals/cohort sizes are explicit, alongside all-observed reviews
Reviewer summaries could duplicate HTTP-attempt cost totals	Only native `api_attempt` records enter cost aggregation; unknown costs/tokens remain visible, with explicit zero-call deterministic components

Finding	Resolution
Repeated denial suppression affected full-escalate, violating review-every-valid-proposal baseline semantics	Suppression now belongs to the scope-aware policy only; full-escalate reviews repeated valid proposals while budget remains; regression expectation corrected
Reviewer logs lacked proposed arguments, stable call identity and exact request attempts	Verdict logs now include identity/arguments/initial prompt; `reviewer_requests.json` preserves exact wire requests including repair/retry requests
Swallowed usage callback errors could produce apparently successful but uncosted episodes	Reviewer usage audit failures and assessor trace-persistence failures mark the episode as an error; result scoring fields become unknown
Executor completed with exit status zero even when episodes failed	CLI returns nonzero for errored selections; analyzer returns nonzero for incomplete/audit-invalid or unmatched/inadequate studies unless explicitly exporting diagnostics
No model-free manifest validation CLI	Added `--validate-only`; no endpoint calls or output-directory creation
Executor could select other domains from an old mixed manifest without an explicit suite flag	Default execution selection is banking; other suite support requires an explicit suite selection
Smoke command defaulted to all suites	Banking is now the smoke default and `--suite banking` is documented
Sanity test expected stock `AgentPipeline` after factory migrated to repeated custom runtime	Test now verifies `BasePipelineElement` compatibility and the actual `GatedAgentPipeline` class
Figure dependency absent from harness requirements; transitive environment not frozen	Added matplotlib and root `requirements-evaluation.lock` containing the tested dependency set
Failure mining and escalation evidence were not connected to native logs	Exports now include failure candidates with trace paths and blank confirmation fields, review-by-tool outcomes, assessment statuses and escalation tables

Verified behavior and commands

Validation was performed in the repository-local Python 3.12 environment installed from the project requirements, then frozen in the evaluation lock.

- Harness regression suite: **194 passed, 1 skipped, 30 subtests passed**. The skipped case is the opt-in live local-model assessor test, not an offline integration failure.
- Root Gemini runner/adapter tests: **5 passed**. These validate their separate stock-runner path, not gated model performance.

- Explicit legacy R5 logger, cost and metric scripts passed. They are not collected as pytest test functions and retain their old schema semantics.
- Banking offline smoke: **10 scripted integration episodes**, five existing baseline/stub policies, clean plus fixed attack. Real environments, tools, attack setup and official evaluators were used; actors and reviewer were scripted. These results are tagged `offline_test` and rejected by benchmark comparison/reporting.
- A full **9,600-row banking-only manifest** was generated and every row validated without model calls. The documented smaller pilot arithmetic and identity selection are also covered by tests.
- Fake-provider integration verified that `hybrid_llm` actually invokes R4, passes no attack-annotation fields, records policy API usage and persists assessor traces. This is transport/schema evidence, not model-quality evidence.
- Native report exports and plots were generated from synthetic schema-test records in a temporary directory. Clean sentinel direction, attacked TCR, both HITL cohorts, missing/error bounds, consultation-summary deduplication, offline-fixture rejection and output-directory protection are regression tested. Chart layout was visually inspected; these synthetic charts are not submitted as benchmark results.
- `git diff --check` passed. Mermaid source and LaTeX display delimiters were checked; diagrams describe current code branches, including advisory R4. No external Mermaid rendering or live model validation is claimed.

Exact setup, tests, smoke, manifest generation, live preflight, pilots, full sweep, resume and export commands are in the guide. Temporary smoke/plot-test artifacts are outside the submission tree and are not empirical project evidence.

Remaining limitations and evidence gaps

1. **Live readiness:** start local Ollama, ensure `qwen3:14b` is available, and pass the matched live pilot. Native tool calling, schema response quality, model latency/context limits and real usage need live validation. The endpoint was unavailable during this audit.
2. **Scope coverage:** exact grants and bounded prerequisite/rent rules are implemented; arbitrary banking natural language remains unresolved and relies on conservative relevance/review handling. This can produce substantial utility loss. It is a limitation of the current original policy, not evidence that all banking tasks have been solved.
3. **Semantic R4:** policy version `r3-r4-review-selection-v1` now uses eligible conflicts for denial and supported scope/operation for review relevance. It cannot create grants or bypass independent review. Model errors can still cause false denials or unnecessary reviews; fresh development pilots and matched held-out runs are required. See the combined-policy specification.
4. **Strict safety metric:** official AgentDojo attack goals do not label every unauthorized call. The current accepted project metric is AgentDojo-based TCR; executed-call audit supplements it. Do not relabel attack-goal failure as proof that every action was authorized.
5. **Experiment design:** freeze budget grid, coefficients, model digest, limits, attacks, task split and retry protocol after development. The guide uses all 16 banking identities descriptively and a separately analyzed 13-task held-out subset. Shared attack/task/repeat dependencies require care with uncertainty and significance claims; the exporter does not produce confidence intervals.
6. **Model provenance:** automatic `config.json` weights digest remains empty. Capture full tags/digest, model file, Ollama version and hardware with the guide's provenance commands.

Runtime hashes alone cannot establish identical model weights.

7. **Empirical deliverables:** full model results, genuine curves, confirmed representative failures, causal/qualitative escalation-value discussion, report and presentation are still to be produced. Machine completeness checks are necessary but do not certify a finished submission.
8. **Manual failure classification:** an attack following reviewer approval must not automatically be called an auto-approval policy false negative. A utility failure alone does not prove over-conservatism. Confirm decisive calls from trajectories; do not invent examples to fill the handout categories.
9. **Local API accounting:** zero API dollars excludes compute/electricity. Report tokens, unknown token counts and component inference overhead. R4 has a single attempt; agent/reviewer default to one transport attempt, with at most one reviewer formatting repair per consultation.
10. **Submission packaging:** the blanket `runs/` ignore rule has been removed so intended benchmark evidence can be tracked. Removing it does not stage or commit files. Preserve full audit/provenance and copy publishable results, figures, curated cases, report and slides into the submitted artifact tree. Old local traces and historical R3 documents are not evidence of the new frozen run.

No push, PR, live full evaluation or external publication was performed by this audit.